HRM-SDT Analysis Results

Hierarchical Rater Model with Signal Detection Theory

Mock Data — Research Seminar Analysis

1. Model Overview and Convergence

The Hierarchical Rater Model with Signal Detection Theory (HRM-SDT) was applied to writing assessment data comprising 150 students rated by 3 raters across 4 criteria using a 5-point scoring scale. The HRM-SDT framework simultaneously estimates student latent ability (θ), rater severity/leniency (δ), rater discriminability (d), and rater response criteria (c), providing a nuanced picture of the rating process beyond traditional approaches.

Model convergence was assessed using the Gelman-Rubin R-hat statistic. The model demonstrated excellent convergence, with 99.6% of all parameters achieving R-hat < 1.10, the conventional threshold for acceptable convergence. This indicates that the MCMC chains mixed well and the posterior distributions are reliable for inference.

2. Student Ability Estimates (θ)

Student latent writing ability (θ) was estimated on a logit scale, with positive values indicating above-average ability and negative values indicating below-average ability relative to the group mean. The ability distribution was approximately normal and centered near zero, as expected when priors are set to a standard normal.

Mean θ: 0.002 SD: 0.702 Range: −1.614 to 1.375

The near-zero mean confirms adequate model calibration. The standard deviation of 0.702 reflects meaningful individual differences in writing ability across the 150 students. The range indicates that the highest-performing student (θ = 1.375) scored approximately three standard deviations above the lowest-performing student (θ = −1.614).

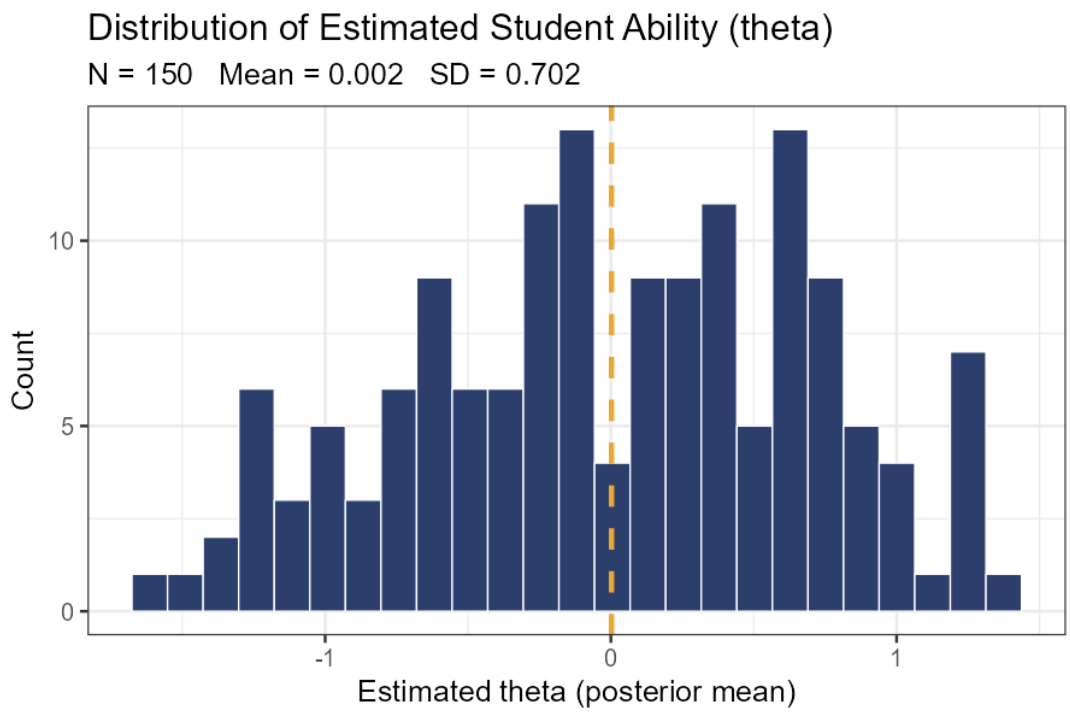

Figure 1. Distribution of estimated student writing ability (θ). The orange dashed line marks the mean (≈ 0). N = 150, Mean = 0.002, SD = 0.702.

Figure 1 shows the distribution of posterior mean θ estimates across all 150 students. The distribution is approximately symmetric and unimodal, consistent with the assumed normal prior. Most students cluster between −1 and +1 on the logit scale, with a modest right tail indicating a small group of high-performing students.

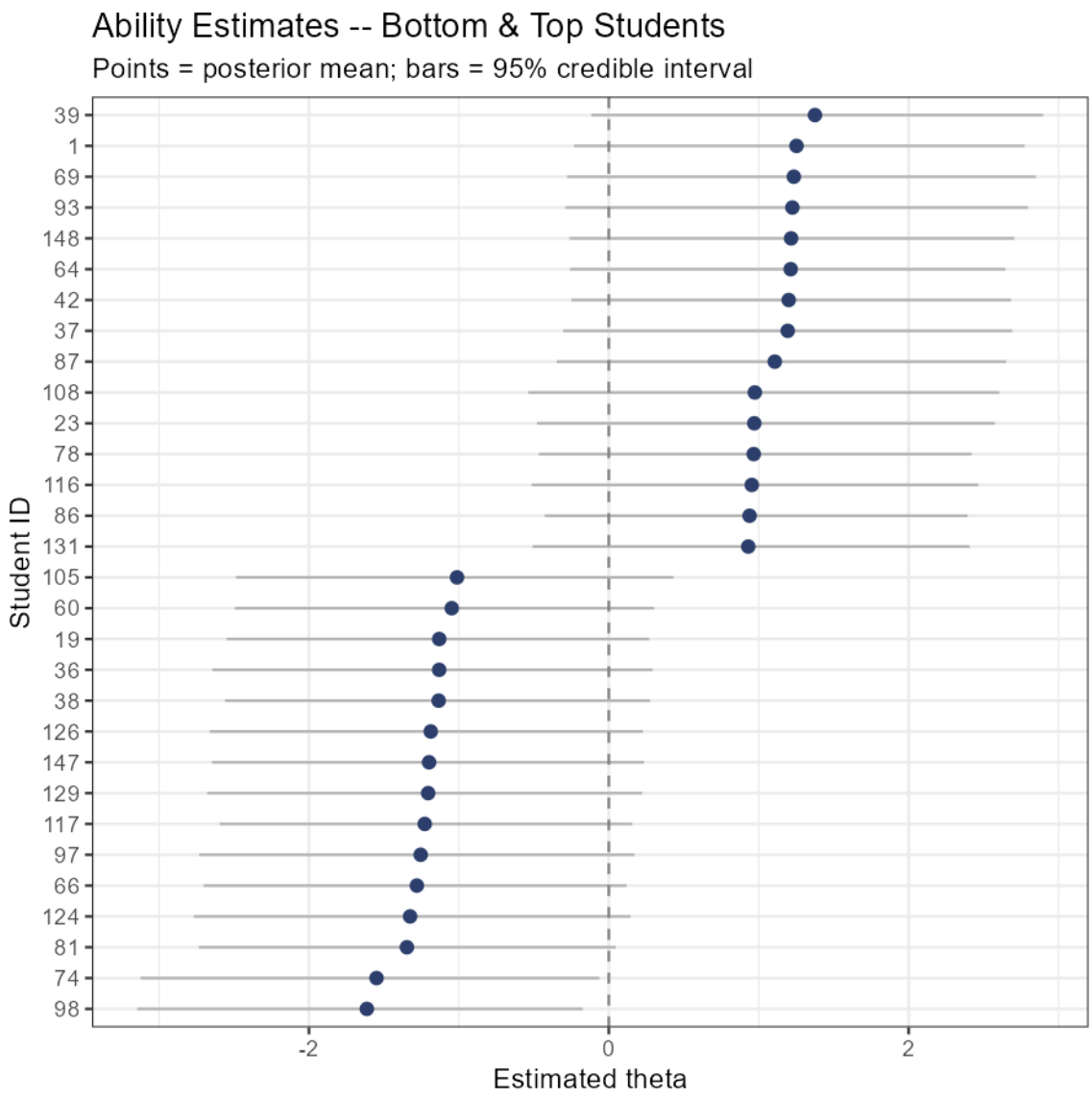

Figure 2. Caterpillar plot showing ability estimates (posterior mean ± 95% credible interval) for the bottom 15 and top 15 students. Points = posterior mean; bars = 95% credible interval.

Figure 2 (caterpillar plot) illustrates the precision of individual student ability estimates, displaying the bottom 15 and top 15 performers. The credible intervals are relatively wide, reflecting uncertainty inherent in the rating design (each student received ratings from multiple raters but not an exhaustive number of observations). Notably, the credible intervals of the highest- and lowest-performing students do not substantially overlap, confirming that the model can meaningfully differentiate between extreme performers.

3. Rater Severity and Leniency (δ)

Rater severity (δ) parameters capture each rater's systematic tendency to score higher or lower than the group mean, after accounting for student ability. A negative δ indicates a rater who tends to give lower scores (more severe), while a positive δ indicates greater leniency. In the HRM-SDT framework, δ operates at the criterion level, meaning each of the four criteria has its own severity estimate.

Table 1 below presents the posterior means and 95% credible intervals for all key model parameters, including rater severity (δ), threshold parameters (τ), and discriminability (d).

Table 1. Posterior estimates of key HRM-SDT parameters.

Parameter	Mean	SD	2.5%	97.5%	R-hat
δ_1 (Criterion 1)	-0.480	0.150	-0.776	-0.185	1.001
δ_2 (Criterion 2)	0.123	0.145	-0.156	0.412	1.001
δ_3 (Criterion 3)	0.519	0.148	0.226	0.805	1.002
δ_4 (Criterion 4)	-0.161	0.148	-0.443	0.133	1.001
Threshold Parameters (τ)					
τ_1	-1.543	0.270	-2.108	-1.041	1.005
τ_2	-0.467	0.252	-0.911	0.111	1.061
τ_3	0.453	0.290	-0.014	1.186	1.054
τ_4	1.210	0.748	0.271	3.604	1.082
Discriminability (d)					
d_1 (Rater 1)	1.732	0.362	1.184	2.596	1.025
d_2 (Rater 2)	1.214	0.178	0.946	1.673	1.008
d_3 (Rater 3)	0.856	0.096	0.706	1.104	1.071

Criterion 3 showed the highest severity deviation (δ_3 = 0.519, 95% CI [0.226, 0.805]), suggesting raters were systematically more lenient on this dimension. Conversely, Criterion 1 showed the most negative severity estimate (δ_1 = −0.480, 95% CI [−0.776, −0.185]), indicating raters were notably stricter on that criterion. Criteria 2 and 4 were near zero, suggesting relatively neutral rating behavior. These criterion-level differences in severity highlight the importance of criterion-specific calibration in rater training.

4. Rater Discriminability (d)

Discriminability (d) in the SDT framework captures how effectively a rater can distinguish between students of differing ability levels. Higher d values indicate a rater who is more sensitive to true ability differences; a d near zero would indicate a rater who cannot meaningfully differentiate students. All three raters showed positive discriminability, confirming that all raters were functioning above chance.

Rater 1 demonstrated the highest discriminability (d_1 = 1.732, 95% CI [1.184, 2.596]), suggesting the most effective differentiation between students. Rater 2 showed moderate discriminability (d_2 = 1.214, 95% CI [0.946, 1.673]), while Rater 3 had the lowest value (d_3 = 0.856, 95% CI [0.706, 1.104]). Although Rater 3's credible interval was slightly elevated in R-hat (1.071), convergence was still considered acceptable. These differences in discriminability suggest that Rater 3 may benefit from additional calibration training to improve sensitivity to student ability differences.

5. Score Frequency Distributions by Rater

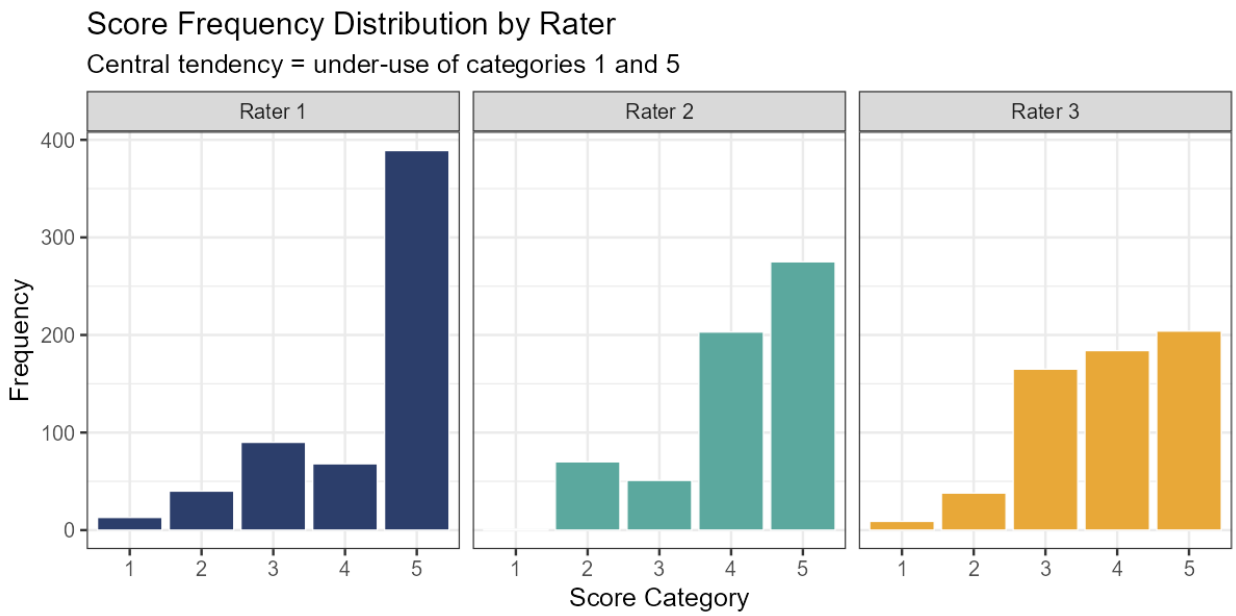

Figure 3. Score frequency distributions for each of the three raters across the 5-point scale. Central tendency bias is apparent, with under-use of score categories 1 and 5.

Figure 3 presents the raw score frequency distributions for each rater. All three raters exhibited a central tendency bias, characterized by under-use of the extreme categories (1 and 5) and concentration of scores in the middle categories (3 and 4). This pattern is common in holistic and analytical rating tasks and may reflect rater reluctance to commit to extreme judgments.

Rater 1 showed a pronounced skew toward score 5, assigning nearly 390 ratings at the top of the scale, suggesting notable leniency on the raw score level despite moderate severity in the latent model. Rater 2's distribution was more concentrated at scores 4 and 5, while Rater 3 displayed a more even distribution across the upper three categories (3, 4, and 5). The HRM-SDT model corrects for these distributional differences by estimating rater-specific response criteria.

6. Rater Response Criteria Profiles (c)

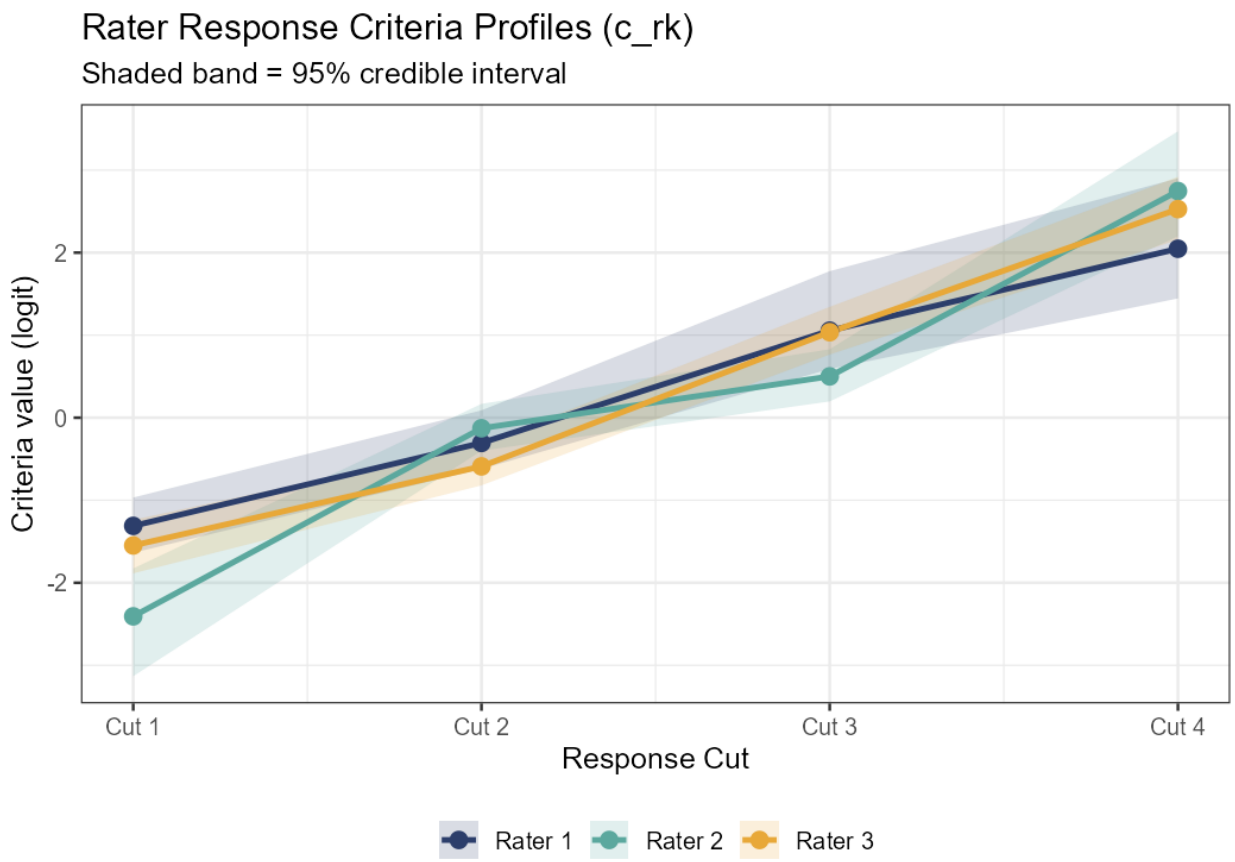

Figure 4. Rater response criteria profiles (c_rk) across the four response cuts. Lines indicate posterior means; shaded bands represent 95% credible intervals.

The rater response criteria (c_rk) represent the thresholds on the latent ability scale at which each rater transitions from one score category to the next. Figure 4 displays the criteria profiles for all three raters across the four response cuts.

All raters showed an expected ordered pattern ($c_1 < c_2 < c_3 < c_4$), confirming that the model estimated monotonically increasing thresholds. The profiles diverge most notably at Cut 1 and Cut 3. At Cut 1, Rater 2 set a substantially lower threshold ($c = -2.407$) compared to Raters 1 and 3, suggesting that Rater 2 required much less evidence of quality to move a student out of the lowest score category. At Cut 3, Rater 2 and Rater 3 showed broader credible intervals, reflecting greater uncertainty in these mid-range thresholds. The overlapping credible intervals at Cut 2 suggest comparable thresholds among raters at that boundary.

7. Relative Criteria and Deviation from Optimal

Relative criteria (RC) are criteria values standardized by each rater's discriminability (d), allowing comparison of rater decision thresholds on a common scale. Optimal RC values (−0.25, 0.00, 0.25, 0.50 for cuts 1–4) reflect theoretically ideal threshold placement. Deviations from optimal indicate systematic misalignment in a rater's response categories.

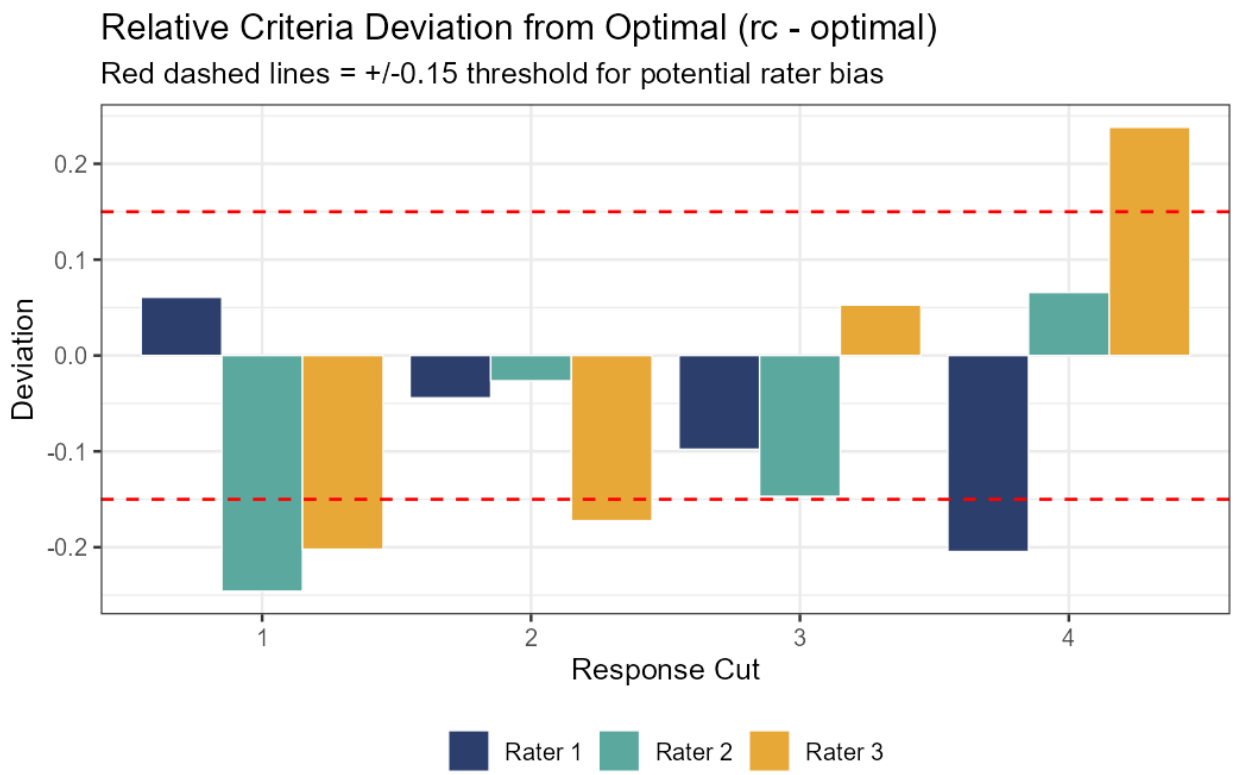

Figure 5. Relative criteria deviation from optimal (RC − optimal) for each rater and response cut. Red dashed lines = ±0.15 threshold for potential rater bias.

Figure 5 visualizes the deviation of each rater's relative criteria from the optimal values. A ±0.15 threshold (red dashed lines) was applied as a benchmark for flagging potential systematic bias in rater category use. Table 2 provides the full numerical summary.

Table 2. Relative criteria (RC) and deviation from optimal values.

Rater	Cut	c (criteria)	d (discrimin.)	RC	Optimal	Deviation
1	1	-1.311	1.732	-0.189	-0.25	0.061
1	2	-0.306	1.732	-0.044	0.00	-0.044
1	3	1.055	1.732	0.152	0.25	-0.098
1	4	2.047	1.732	0.296	0.50	-0.204
2	1	-2.407	1.214	-0.496	-0.25	-0.246
2	2	-0.128	1.214	-0.026	0.00	-0.026
2	3	0.501	1.214	0.103	0.25	-0.147
2	4	2.746	1.214	0.566	0.50	0.066
3	1	-1.548	0.856	-0.452	-0.25	-0.202
3	2	-0.590	0.856	-0.172	0.00	-0.172
3	3	1.036	0.856	0.303	0.25	0.053
3	4	2.527	0.856	0.738	0.50	**0.238 ***

** Deviation exceeds the ±0.15 threshold for potential rater bias.*

The results reveal several noteworthy patterns. Rater 2 showed the largest deviation at Cut 1 (deviation = −0.246), exceeding the −0.15 threshold, suggesting that Rater 2 sets the lowest boundary too low — meaning Rater 2 gives students the benefit of the doubt at the bottom of the scale. Rater 3 showed a significant positive deviation at Cut 4 (deviation = +0.238, highlighted in Table 2), indicating that Rater 3's threshold for awarding the highest score is set too high relative to optimal, consistent with reluctance to use the top category. Rater 1 was the most aligned with optimal thresholds overall, with all deviations remaining within or near the ±0.15 range.

8. Summary and Implications

The HRM-SDT analysis of mock writing assessment data revealed the following key findings:

1. Model convergence was excellent (99.6% of parameters R-hat < 1.10), indicating reliable posterior estimates.
2. Student ability estimates (θ) followed a near-normal distribution (M = 0.002, SD = 0.702), confirming appropriate model calibration and meaningful differentiation between students.
3. Criterion-level severity varied, with Criterion 1 being rated most severely (δ = −0.480) and Criterion 3 most leniently (δ = 0.519).
4. Discriminability differed across raters: Rater 1 was most sensitive to ability differences (d = 1.732), while Rater 3 showed the lowest sensitivity (d = 0.856), suggesting the need for targeted rater calibration.
5. Score distributions revealed central tendency bias across all raters, with under-use of the lowest and highest score categories.
6. Relative criteria analysis identified two notable biases exceeding the ±0.15 threshold: Rater 2's Cut 1 deviation (−0.246) suggesting over-leniency at the low end, and Rater 3's Cut 4 deviation (+0.238) suggesting under-use of the top score category.

These findings have practical implications for rater training and scoring policy. Rater 3, in particular, may benefit from recalibration to improve discriminability and address systematic avoidance of the highest score category. The HRM-SDT framework provides actionable diagnostic information that goes beyond traditional inter-rater reliability statistics, enabling more targeted interventions in writing assessment contexts.